Resonance-Manifold Solutions in SEFI

Jason Dutton

Overview

Resonance manifolds describe combined static and dynamic curvature behavior in the SEFI metric.

Dynamic Resonance Component

$$\Psi_{\mu\nu}(x,t) = A_{\mu\nu}\sin(\omega t - kx)$$

Static Curvature Component

$$\Phi_{\mu\nu}(x) = B_{\mu\nu}e^{-kx}$$

Combined Resonance Manifold

$$g_{\mu\nu} = \eta_{\mu\nu} + B_{\mu\nu}e^{-kx} + A_{\mu\nu}\sin(\omega t - kx)$$

Interpretation

The SEFI resonance manifold unifies decaying static curvature with oscillatory dynamic curvature in a single geometric structure.